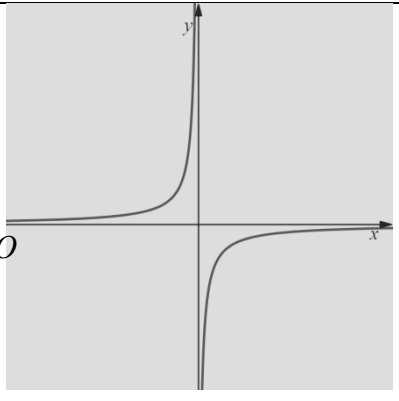
1. (a)	$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \Rightarrow \frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2x^2}$	M1 A1 A1 **(3)**
(b)	$x = \frac{1}{2} \Rightarrow y = 3$	B1
	Substitutes $x = \frac{1}{2}$ into their $\frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2x^2} = ...\left(= \frac{5}{2}\right)$	M1
	Uses the perpendicular gradient rule Eg. $\frac{5}{2} \rightarrow -\frac{2}{5}$	dM1
	Attempts the equation of the normal at P $y - 3 = -\frac{2}{5}\left(x - \frac{1}{2}\right)$	M1
	$2x + 5y - 16 = 0$ oe	A1 **(5)**
		(8 marks)

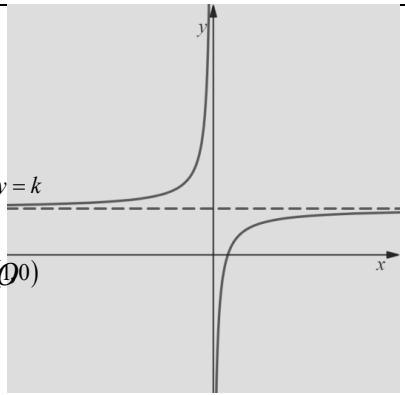
2 .(a)	Attempts to use the given model at least once. Eg $2^3 = p \times 3^2 + q$	M1
	Two correct simplified equations $9p + q = 8$ $25p + q = 13.8(24)$	A1
	Solves simultaneously to get at least one of p or q	dM1
	$p = 0.364, q = 4.72(4)$	A1 **(4)**
(b)	Attempts to find T when $H = 5$ Eg. Calculates $\sqrt{\frac{125 - "q"}{"p"}}$	M1
	$(T =)$ 18.2	A1 **(2)**
		(6 marks)

3. (a)(i)	$P(-180,-4)$	B1, B1
(ii)	$Q(450,0)$	B1 **(3)**
(b)	$R(360,7)$	B1, B1 **(2)** **(5 marks)**

4.(a)	Gradient $PQ = -3$	B1
	Attempts to find equation of l Eg. $y-13=-3(x+2)$	M1
	$y=-3x+7$	A1
		(3)
(b)	Attempts to use minimum is $(4,-5)$ Eg $y=...(x-4)^2-5$	M1
	Attempts to use $(-2,13)$ with $y=a(x-4)^2-5 \Rightarrow a=...$	dM1
	$\Rightarrow y=\frac{1}{2}(x-4)^2-5$ or $y=\frac{1}{2}x^2-4x+3$ oe	A1
		(3)
(c)	Two of $y>-3x+7$, $y<\frac{1}{2}(x-4)^2-5$ $x<-2$	M1
	All three of $y>-3x+7$, $y<\frac{1}{2}(x-4)^2-5$ $x<-2$	A1
		(2)
		(8 marks)

5 (a)	Attempts the sine rule $\frac{\sin\alpha}{14}=\frac{\sin 0.43}{6}$	M1
	$\Rightarrow \alpha = 1.337$ (radians) Accept awrt 1.33/1.34 or awrt 76.6/76.7 ($^\circ$)	A1
	angle $AOD = \pi - 1.337 =$ awrt 1.805 (radians)	A1 **(3)**
(b)	Attempts $s = r\theta$ with $r = 6$ and an allowable θ	M1
	Arc length ABC = awrt 26.9 m	A1 **(2)**
(c)	Attempts $\frac{1}{2}r^2\theta$ with $r = 6$ and an allowable θ in radians (= 80.6)	M1
	Attempts area $AOD = \frac{1}{2}\times 6\times 14\times \sin("0.91")$ oe (= 33.1)	M1
	Attempts sector + triangle with correct attempt at angles	dM1
	= 113.7 (m^2)	A1 **(4)** **(9 marks)**

6.(a)		Negative reciprocal shape Fully correct	M1 A1 **(2)**

(b)		Graph is part (a) translated ↑	B1ft
		Correct asymptote or intercept	B1
		Correct asymptote and intercept	B1 **(3)**
(c)	Sets $3x+4=-\frac{k}{x}+k \Rightarrow 3x^2+(4-k)x+k=0$		M1, A1
	Attempts use $b^2-4ac=0$ to find the critical values		M1
	Uses $b^2-4ac<0$ and selects inside region for critical values		dM1
	$10-2\sqrt{21}<k<10+2\sqrt{21}$		A1 **(5)** **(10 marks)**

7. (a)	$2x-3\sqrt{x}-5=9 \Rightarrow 2x-3\sqrt{x}-14=0$ and treats as quadratic equation	M1	
	$\Rightarrow \left(2\sqrt{x}-7\right)\left(\sqrt{x}+2\right)=0 \Rightarrow \left(\sqrt{x}=\right)\frac{7}{2},(-2)$	A1	
	$\Rightarrow x=\left(\frac{7}{2}\right)^2=\frac{49}{4}$	dM1 A1	
			(4)
(b)	$\left(\mathrm{f}'(x)=\right)\ 2-\frac{3}{2}x^{-\frac{1}{2}}$	B1	

	$\left(\mathrm{f}''(x)=\right)\ \frac{3}{4}x^{-\frac{3}{2}}$	M1 A1
	Attempts $\frac{3}{4}x^{-\frac{3}{2}}=6\Rightarrow x^{-\frac{3}{2}}=8\Rightarrow x=\frac{1}{4}$	dM1 A1 **(5)** **(9 marks)**

8 (a)	$x > 4$	B1 **(1)**
(b)	$(3x-2)^2(x-4) = (9x^2-12x+4)(x-4)$ $= 9x^3 - 48x^2 + 52x - 16$	M1 A1 A1 **(3)**

(c)	Sets $9x^3 - 48x^2 + 52x - 16 = -16 \Rightarrow 9x^3 - 48x^2 + 52x\ (= 0)$	B1ft
	$\Rightarrow x\left(9x^2 - 48x + 52\right) = 0 \Rightarrow x = \dfrac{48 \pm \sqrt{48^2 - 4 \times 9 \times 52}}{18} = \dfrac{16 \pm 4\sqrt{3}}{6}$	M1 B1
	Distance $PQ = \dfrac{16 + 4\sqrt{3}}{6} - \dfrac{16 - 4\sqrt{3}}{6} = \dfrac{4}{3}\sqrt{3}$	M1 A1
		(5) **(9 marks)**

9. (i)	$\dfrac{(3x+2)^2}{4\sqrt{x}} = \dfrac{9x^2+12x+4}{4\sqrt{x}} = \dfrac{9}{4}x^{\frac{3}{2}} + 3x^{\frac{1}{2}} + x^{-\frac{1}{2}}$	M1
	$\displaystyle\int \dfrac{(3x+2)^2}{4\sqrt{x}}\,\mathrm{d}x = \dfrac{2}{5}\times\dfrac{9}{4}x^{\frac{5}{2}} + \dfrac{2}{3}\times 3x^{\frac{3}{2}} + 2\times x^{\frac{1}{2}}(+c) = \dfrac{9}{10}x^{\frac{5}{2}} + 2x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c$	dM1 A1 A1 A1
		(5)
(ii)	$\mathrm{f}'(x) = x^2 + ax + b$	
	Attempts to use $\mathrm{f}'(3) = 2 \Rightarrow$ $2 = 9 + 3a + b$	M1
	Attempts to integrate $\left(\mathrm{f}(x) =\right) \dfrac{1}{3}x^3 + \dfrac{1}{2}ax^2 + bx + c$	M1
	Attempts to use y intercept $= -8$ and $(3,-2)$ in $\mathrm{f}(x) = \dfrac{1}{3}x^3 + \dfrac{1}{2}ax^2 + bx + c$	dM1
	Correct equation in a and b $\quad -2 = 9 + \dfrac{9}{2}a + 3b - 8$	A1
	Solves simultaneously to get values for a and b	ddM1
	$a = -4, b = 5 \Rightarrow \left(\mathrm{f}(x) =\right) \dfrac{1}{3}x^3 - 2x^2 + 5x - 8$	A1
		(6)
		(11 marks)